Solutions:

1. D. 15x – 10y – 9 = 0

Using slope-intercept form:
$y = mx + b$

$$3x - 2y + 6 = 0$$
$$y = \frac{3}{2}x + 3$$

By inspection, $m_1 = 3/2$
Since parallel: $m_2 = m_1$
From the choices,

$$15x - 10y - 9 = 0$$
$$y = \frac{3}{2}x - \frac{9}{10}$$

By inspection, $m_2 = 3/2$

2. C. 4x + 5y = 2

Using two-point form:

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$
$$y + 2 = \frac{6+2}{-7-3}(x-3)$$
$$y + 2 = \frac{8}{-10}(x-3)$$
$$-10y - 20 = 8x - 24$$
$$8x + 10y - 4 = 0$$
$$4x + 5y - 2 = 0$$

3. B. 4

$$\text{Given line 1}: 4x - 3y - 12 = 0$$
$$A = 4; B = -3; C_1 = -12$$
$$\text{Given line 2}: 4x - 3y + 8 = 0$$
$$A = 4; B = -3; C_2 = 8$$

$$d = \frac{C_2 - C_1}{\sqrt{A^2 + B^2}}$$
$$d = \frac{8-(-12)}{\sqrt{4^2 + (-3)^2}}$$
$$d = \frac{20}{5}$$
$$d = 4 \text{ units}$$

4. D. (-5/3, 1)

$$3x + 2y^2 - 4y + 7 = 0$$
$$y^2 - 2y + \frac{3}{2}x + \frac{7}{2} = 0$$

By completing square:

$$y^2 - 2y = -\frac{3}{2}x - \frac{7}{2}$$
$$(y-1)^2 = -\frac{3}{2}x - \frac{7}{2} + (1)^2$$
$$(y-1)^2 = -\frac{3}{2}x - \frac{5}{2}$$
$$(y-1)^2 = -\frac{3}{2}\left(x + \frac{5}{3}\right)$$

Standard equation:
$$(y-k)^2 = -4a(x-h)$$

By inspection, h = - 5/3 and k = 1, thus the vertex is at (-5/3, 1)

5. A. 2, 8

$$y^2 + 8x - 6y + 25 = 0$$
$$y^2 - 6y = -8x - 25$$

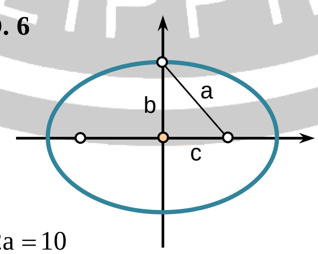
By completing square:

$$(y-3)^2 = -8x - 25 + (3)^2$$
$$(y-3)^2 = -8x - 16$$
$$(y-3)^2 = -8(x+2)$$

Standard equation:
$$(y-k)^2 = -4a(x-h)$$

By inspection, 4a = 8 thus, a = 2
Focal length = a = 2
Length of latus rectum = 4a = 8

6. D. 6



$$2a = 10$$
$$a = 5$$
$$2b = 8$$
$$b = 4$$

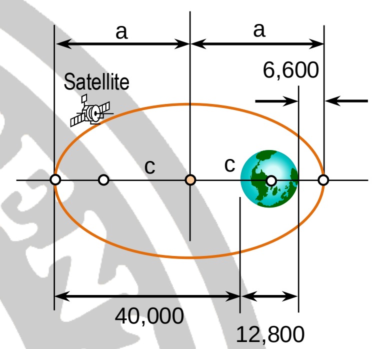
Solving for c:

$$c = \sqrt{a^2 - b^2}$$
$$c = \sqrt{5^2 - 4^2}$$
$$c = 3$$

Distance between foci = 2c

$$2c = 2(3)$$
$$2c = 6$$

7. D. 0.56



$$2a = 40,000 + 12,800 + 6,600$$
$$a = 29,700$$
$$c = a - (6,600 + 6,400)$$
$$c = 29,700 - (6,600 + 6,400)$$
$$c = 16,700$$
$$e = \frac{c}{a}$$
$$e = \frac{16,700}{29,700}$$
$$e = 0.56$$

8. A. 1.80

$$9x^2 - 4y^2 - 36x + 8y = 4$$
$$9(x^2 - 4x) - 4(y^2 - 2y) = 4$$

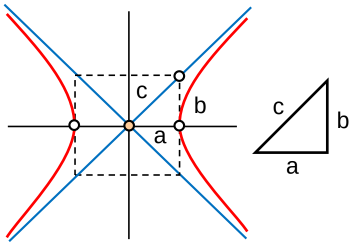
By completing square:

$$9(x-2)^2 - 4(y-1)^2 = 4 + 9(2)^2 - 4(1)^2$$
$$9(x-2)^2 - 4(y-1)^2 = 36$$
$$\frac{(x-2)^2}{4} - \frac{(y-1)^2}{9} = 1$$

Standard equation:

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

By inspection, $a^2 = 4$ thus, a = 2.
Also, $b^2 = 9$ thus, b = 3.



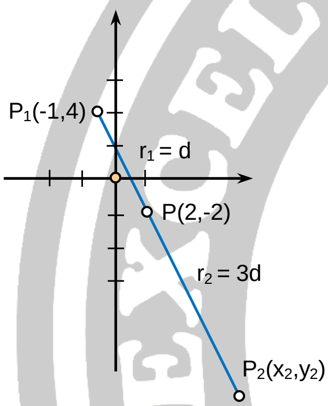
Solving for c

$$c = \sqrt{a^2 + b^2} = \sqrt{4+9} = 3.605$$

Solving for e:

$$e = \frac{c}{a} = \frac{3.605}{2} = 1.8$$

9. D. (11, -20)



$$x = \frac{x_1 r_2 + x_2 r_1}{r_1 + r_2}$$

$$2 = \frac{(-1)(3d) + x_2(d)}{d + 3d}$$

$$2 = \frac{-3\cancel{d} + \cancel{d}x_2}{4\cancel{d}}$$

$$x_2 = 11$$

$$y = \frac{y_1 r_2 + y_2 r_1}{r_1 + r_2}$$

$$-2 = \frac{4(3d) + y_2(d)}{d + 3d}$$

$$-2 = \frac{12\cancel{d} + \cancel{d}y_2}{4\cancel{d}}$$

$$y_2 = -20$$

10. C. 45°

$$2x + y - 8 = 0$$

$$y = -2x + 8$$

$$m_1 = -2$$

$$x + 3y + 4 = 0$$

$$y = -\frac{1}{3}x - \frac{4}{3}$$

$$m_2 = -1/3$$

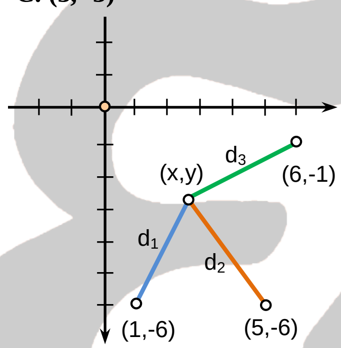
Let:

θ = angle between the two lines

$$\theta = \tan^{-1}\left(\frac{m_2 - m_1}{1 + m_1 m_2}\right)$$

$$\theta = \tan^{-1}\left(\frac{-1/3 - (-2)}{1 + (-2)(-1/3)}\right) = 45^\circ$$

11. C. (3, -3)



$$d_1 = d_2$$

$$\sqrt{(x-1)^2 + (y+6)^2} = \sqrt{(x-5)^2 + (y+6)^2}$$

$$(x-1)^2 + \cancel{(y+6)^2} = (x-5)^2 + \cancel{(y+6)^2}$$

$$\cancel{x^2} - 2x + 1 = \cancel{x^2} - 10x + 25$$

$$8x = 24$$

$$x = 3$$

$$d_2 = d_3$$

$$\sqrt{(x-5)^2 + (y+6)^2} = \sqrt{(x-6)^2 + (y+1)^2}$$

$$(x-5)^2 + (y+6)^2 = (x-6)^2 + (y+1)^2$$

Substitute x = 3:

$$(x-5)^2 + (y+6)^2 = (x-6)^2 + (y+1)^2$$

$$(3-5)^2 + (y+6)^2 = (3-6)^2 + (y+1)^2$$

$$4 + \cancel{y^2} + 12y + 36 = 9 + \cancel{y^2} + 2y + 1$$

$$10y = -30$$

$$y = -3$$

Thus the point is at (3, -3)

12. A. 2x – 3y = 0

$$\frac{x^2}{9} - \frac{y^2}{4} = 1$$

Standard equation: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

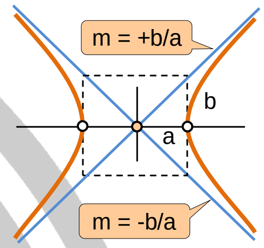
By inspection:

$$a^2 = 9$$

$$a = 3$$

$$b^2 = 4$$

$$b = 2$$



Equation of the asymptotes of a hyperbola with center at (0,0):

$$y = \pm\frac{b}{a}x$$

$$y = \pm\frac{2}{3}x$$

$$2x - 3y = 0 \text{ or}$$

$$2x + 3y = 0$$

13. B. 2.7

$$4x^2 + 9y^2 + 8x - 32 = 0$$

$$4(x^2 + 2x) + 9y^2 = 32$$

By completing square:

$$4(x+1)^2 + 9y^2 = 32 + 4(1)^2$$

$$4(x+1)^2 + 9y^2 = 36$$

$$\frac{(x+1)^2}{9} + \frac{y^2}{4} = 1$$

Standard equation:

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

By inspection, $a^2 = 9$ thus, $a = 3$.

Also, $b^2 = 4$ thus, $b = 2$.

$$LR = \frac{2b^2}{a} = \frac{2(4)}{3} = 2.667$$

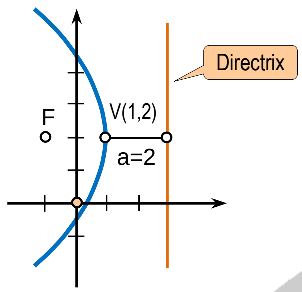
14. B. x = 3

$$(y-2)^2 = -8(x-1)$$

Standard equation:

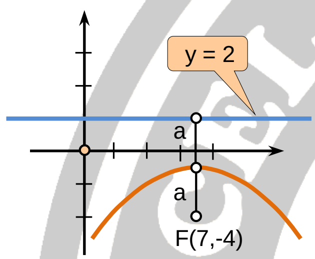
$$(y-k)^2 = -4a(x-h)$$

By inspection, the vertex is at (1, 2). Also 4a = 8, thus a = 2.



From figure, the equation of the directrix is, x = 3.

15. B. $x^2 - 14y + 12x + 61 = 0$



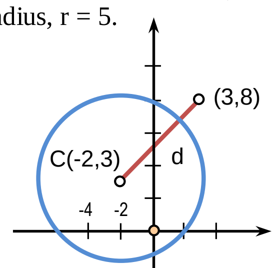
By inspection, 2a = 6 and a = 3
The coordinates of the vertex is at (7,-1).

Standard equation:
$$(x-h)^2 = -4a(y-k)$$

Substituting:
$$(x-7)^2 = -4(3)(y+1)$$
$$x^2 - 14x + 49 = -12y - 12$$
$$x^2 - 14x + 12y + 61 = 0$$

16. A. 2.1

$$x^2 + y^2 + 4x - 6y = 12$$
By completing square:
$$(x+2)^2 + (y-3)^2 = 12 + (2)^2 + (3)^2$$
$$(x+2)^2 + (y-3)^2 = 25 \text{ or } (5)^2$$
By inspection, the center is at (-2, 3) and the radius, r = 5.



Solving for distance between (-2,3) and (3,8):

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$d = \sqrt{(3+2)^2 + (8-3)^2}$$
$$d = 7.071$$

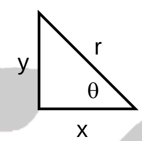
Let: x = shortest distance
$$x = d - r = 7.071 - 5 = 2.071$$

17. C. r = 6 cos θ

Standard equation:
$$(x-h)^2 + (y-k)^2 = r^2$$

Substitute coordinates of the center and radius as given:
$$(x-3)^2 + (y-0)^2 = 3^2$$
$$x^2 - 6x + 9 + y^2 = 9$$
$$x^2 + y^2 = 6x$$

Refer to the triangle:



$$x^2 + y^2 = r^2$$
$$x = r\cos\theta$$

$$x^2 + y^2 = 6x$$
$$r^{\cancel{2}} = 6(\cancel{r}\cos\theta)$$
$$r = 6\cos\theta$$

18. A. 47.1

Note: This is an equation of an ellipse

$$9x^2 + 25y^2 = 225$$
$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$
$$\frac{x^2}{(5)^2} + \frac{y^2}{(3)^2} = 1$$

Standard equation: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

By inspection, a = 5 and b = 3

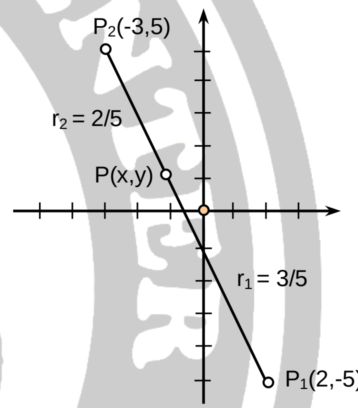
$$A = \pi ab$$
$$A = \pi(5)(3)$$
$$A = 47.12 \text{ square units}$$

19. B. y = -x - 2

$$P_1\ (1,-3) \rightarrow x_1 = 1 \text{ and } y_1 = -3$$
$$P_2\ (-4,2) \rightarrow x_2 = -4 \text{ and } y_2 = 2$$

Using two point form:
$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$
$$y + 3 = \frac{2+3}{-4-1}(x-1)$$
$$y + 3 = -1(x-1)$$
$$y + 3 = -x + 1$$
$$y = -x - 2$$

20. A. (-1,1)



$$x = \frac{x_1 r_2 + x_2 r_1}{r_1 + r_2}$$
$$x = \frac{2(2/5) + (-3)(3/5)}{(3/5) + (2/5)}$$
$$x = -1$$

$$y = \frac{y_1 r_2 + y_2 r_1}{r_1 + r_2}$$
$$y = \frac{-5(2/5) + 5(3/5)}{(3/5) + (2/5)}$$
$$y = 1$$

21. A. 3

Given line $\rightarrow 2x - 3y + 6 = 0$

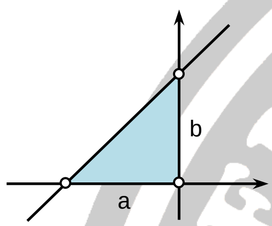
At x = 0:

$$2x - 3y + 6 = 0$$
$$2(0) - 3y + 6 = 0$$
$$y = 2$$

At y = 0:

$$2x - 3y + 6 = 0$$
$$2x - 3(0) + 6 = 0$$
$$x = -3$$

Thus the x-intercept (a) is 3 units and the y-intercept (b) is 2 units.



$$A = \frac{1}{2}ab$$

$$A = \frac{1}{2}(3)(2)$$

$$A = 3 \text{ sq. units}$$

22. D. hyperbola

23. C. Parabola

24. B. 3x – y + 1 = 0

As given: $m = 3;\ b = 1$

Using point slope form:

$$y = mx + b$$
$$y = 3x + 1$$
$$3x - y + 1 = 0$$

25. A. 2.5 units

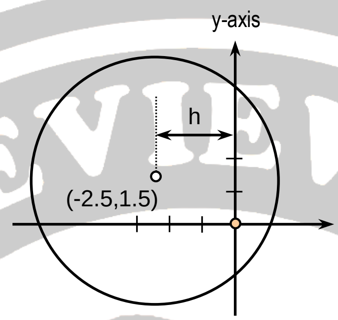
$$2x^2 + 2y^2 + 10x - 6y - 55 = 0$$
$$x^2 + y^2 + 5x - 3y - 27.5 = 0$$

By completing square:

$$\begin{bmatrix} (x+2.5)^2 \\ +(y-1.5)^2 \end{bmatrix} = \begin{bmatrix} 27.5 + (2.5)^2 \\ +(1.5)^2 \end{bmatrix}$$

$$(x+2.5)^2 + (y-1.5)^2 = 36$$

Standard equation: $(x – h)^2 + (y – k)^2 = r^2$
By inspection, h = - 2.5 and k = 1.5



Note: The distance of the center of the circle from the y-axis is equal to h.
Thus, the answer is 2.5 unit